19. RAMS: Reliability, Availability, Maintainability, Safety

19.1. Reliability, Availability and Maintainability: General

19.1.1. The Contractor shall submit Reliability, Availability and Maintainability Management Plan as specified in chapter 2 of the Employer's Requirements, General Specification. The Contractor shall verify, after system design have been completed, that the reliability, availability and maintainability requirement will be met.

19.1.2. The Contractor shall comply with the guidelines of IEC 60300-1, IEC 60300-2 and IEC 60571 for electronic equipment, and IEC 60300-3-5 in meeting the reliability, availability and maintainability requirements of equipment.

19.2. Reliability Requirements

19.2.1. Kolkata Metro Rail Corporation attaches the greatest importance to the attainment of the highest possible Reliability during service of all the equipment and systems supplied and installed under this contract. The design, manufacture, installation and commissioning of the equipment as also the training of the operating and maintenance staff shall be such as to ensure near Zero Failure performance in the initial stages and that the few defects and deficiencies that may be exposed during the Service Trial and the initial reliability growth period of one year are totally eliminated in the bulk supply.

19.2.2. The Contractor shall demonstrate by quantitative methods achievement of the specified levels of reliability for the train and specific individual items of equipment.

19.2.3. An evolving reliability model consisting of reliability block diagrams and probability of success equations shall be developed and submitted to the Engineer for acceptance. This model shall show the relationships required for system and equipment to operate successfully. The reliability block diagrams shall include all elements essential to the successful performance of the system and the interrelationships and interface of these elements.

19.2.4. Reliability apportionment and prediction analysis shall be in accordance with established techniques or standards, which will be submitted for acceptance by the Engineer. The analysis shall provide predictions for each major equipment and sub-system. Predictions shall be based on actual revenue service results for identical equipment operating under service conditions and duty cycles equivalent to KMRCL, or more severe. The analysis shall be carried out in parallel with the design of the train. The relevant apportionment and prediction figures shall be part of the design submission documents for the individual equipment, sub-system and system.

19.2.5. Reliability Apportionment and Prediction Report shall be completed prior to build commencing and reports shall be submitted at this stage for acceptance by the Engineer, who reserves the right to require the Contractor to carry out field data collection to verify the reliability model.

19.2.6. Definitions :

(i) Relevant Failure : A relevant failure of an item is an independent failure which results in a loss of function of that item caused by any of the following:

- A fault in an equipment or sub-system while operating within its design and environmental specification limits;

- Improper operation, maintenance, or testing of the item as a result of the Contractor supplied documentation.

Failures of transient nature including those with post investigation status as 'No fault found', shall be considered as relevant failure if in the opinion of the Engineer these are attributable to rolling stock. The decision of the Engineer shall be final.

(ii) Non-relevant Failure: Any failure of an item not included in the definition of relevant failure, such as the following:

- A failure caused by malfunction of other equipment or subsystem that are not supplied by the Contractor;
- A failure caused by human error, except as noted in Relevant Failure above;
- A failure caused by accidents not associated with the normal operation of the item. Such as collision or striking a foreign object on the right of way;
- A failure caused by operating the equipment or sub-system outside of design or environmental specification limits.

(iii) Service Failure: Any relevant failure or combination of relevant failures during revenue service operations, simulated revenue operations, or during pre-departure equipment status checkouts to determine availability for revenue service, which results in one of the following:

- Non-availability of the train to start revenue service after successful completion of pre-departure checkout;
- Withdrawal of the train from revenue services;
- A delay equivalent to or exceeding 3 minutes.

The discretion of declaring a metro train as Not-available to start revenue service after successful completion of pre-departure checkout or withdrawing a train from revenue service on account of any relevant failure rests solely with the Employer and shall be final.

(iv) Pattern Failure: Repeated occurrence of three or more relevant failures of the same replaceable part, item or equipment in same manner in identical or equivalent applications when they occur at a rate, which is inconsistent with the predicted failure rate of the part, item or equipment. The percentage predicted rate shall be as under:

(a) three or more relevant Service Failures of the same replaceable part, item or equipment in same manner in identical or equivalent applications occur at a rate which is at least 20 % higher than the predicted failure rate of the part, item or equipment and / or

(b) at least 20 % of the same replaceable part, item or equipment in the fleet has a relevant failure in the same manner in identical or equivalent applications during a moving 18 months window starting when the reliability demonstration starts and ending at the end of the Defect Liability Period, these failures will be classed as pattern failures.

(v) Mean Distance Between Failure (MDBF): The MDBF is the ratio of the total operating distance accumulated by the total available fleet of the trains to the total number of Service Failures

(vi) Mean Distance Between Component Failures (MDBCF): The MDBCF of a system is the ratio of the total operating distance accumulated by the total population of identical items in the available fleet of the trains to the total number of Service failures occurring within the population identical items.

19.2.7. Reliability Targets:

(i) Reliability shall be monitored for the respective fleet for 6-car metro trains.

(ii) The fleet average levels of MDBF, as specified in table 19.1, shall be achieved.

Table 19.1: Reliability Targets

Duration	Minimum fleet average MDBF
	6-Car fleet
After 6 months of start of revenue service	60,000
After 12 months of start of revenue service	100,000

Any train shall be counted as available for reliability calculations only after a stabilization period of 6 months after putting the train into revenue service.

(iii) The achieved level of MDBCF of major systems shall be as proposed by the Contractor in the bid. The Tenderer shall submit MDBCF of the major systems as listed in table 19.2 along with the bid.

Table 19.2: MDBCF of major systems

S.N	System / Equipment	MDBCF (km)
(i)	Propulsion System	
	a) Shoe collectors	
	b) HSCB and Earthing switch	
	c) Traction Inverter	
	d) Traction Motor	
(ii)	Auxiliary Supply System	
	a) Auxiliary Converter	
	b) Battery Charger	
	c) Back-up Batteries	
(iii)	Air Supply and Friction Brake Equipment	
(iv)	Door System and Controls	
(v)	VAC System	
(vi)	Communication System	
(vii)	Couplers and Draft Gear	
	a) Automatic couplers	
	b) Semi permanent couplers	
(viii)	Bogies	
	a) Drive gear and coupling	
	b) Primary suspension	
	c) Secondary suspension	
(ix)	Lighting System	
(x)	DMS	

19.3. Reliability Demonstration

19.3.1. The Contractor shall be required to establish a personal computer based Failure Reporting and Corrective Action (FRACAS) System to demonstrate compliance with specified train and equipment reliability. The reliability demonstration of each train shall start after six months of that train in revenue service and shall continue till the end of the defects liability period. Reliability of the trains and of the identified major systems shall be demonstrated on fleet basis.

19.3.2. The Employer shall collect and maintain data on every Service Failure along with the TMS data indicating the probable failure. MDBF and MDBCF shall be calculated throughout the monitoring period. The Contractor shall collect all the relevant details from the Engineer and submit monthly Reliability Demonstration Reports.

19.3.3. In case the Contractor is not able to achieve specified/provided reliability target of MDBF/MDBCF, the Contractor shall take necessary corrective measures either by way of change of design of the relevant equipment/ component or software modification.

19.3.4. The Contractor shall analyze each and every failure/defect of components of various equipments to determine the cause of failure and to propose corrective measures, which would be reviewed by the Engineer.

19.3.5. A record shall be maintained for each and every defect/failure in accordance with FRACAS to be submitted by the Contractor and approved by the Engineer.

19.3.6. Reliability shall be monitored during revenue service operation of the trains. The Contractor shall collect and collate data on each and every deficiency and failure observed by both himself and the Engineer, from handing over the first train to the end of the Defect Liability Period. Each and every failure, whether of component, sub-system or system, during this period shall be subject to a failure analysis to determine the cause of failure? The Contractor shall submit investigation reports for review of the Engineer.

19.3.7. Correction shall be made to components or subsystems that either fail to attain predicted reliability levels or show Pattern Failure, at no additional cost to the Employer.

19.4. Availability Requirements

19.4.1. Availability shall be assessed by the following measure:

Percentage Availability = $$\left[\frac{1-[DT(OPM)+DT(CM)]}{TotalTime}\right]\times 100$$

Where:

(i) Total Time is the time in hours in the assessment period multiplied by the number of trains commissioned under the Contract.

(ii) DT (OPM), or Down Time due to Other Preventive Maintenance, is the total down time in hours due to Preventive Maintenance other than service checks, summed over all sessions carried out on all trains commissioned under the Contract during the assessment period. The trains shall not be due for major overhauls at the time of demonstration and shall therefore be excluded from the assessment.

(iii) DT (CM), or Down Time due to Corrective Maintenance, is the total down time in hours due to corrective maintenance, summed over all sessions carried out on the trains commissioned under the Contract during the assessment period. Any unreasonable delay in handing-over the train for repairs for reasons not attributable to

contractor shall be excluded. Time spent on train integrity inspections after train reformations arising from corrective maintenance work shall be included.

(iv) The down time DT (OPM) shall be counted starting from the moment when the train becomes unfit for service or work is physically started on a train, whichever is earlier, and shall end when the train is restored to service condition. If the train is withdrawn from revenue service specially for preventive maintenance, time spent on withdrawing the train and sending back the train to revenue service, if any, shall also be included.

(v) Down time DT (CM) shall be counted starting from the moment when the train becomes unfit for service or work is physically started on a train, whichever is earlier, and shall end when the train is restored to service condition. If the train is sent to revenue service after the corrective maintenance, the time spent on sending back the train to revenue service, if any, shall also be included.

(vi) The down times DT (OPM) and DT (CM) shall also cover the full content of the maintenance work concerned, including safety precautions, inspections, servicing, replacement of equipment, defect detection and rectification, testing and restoration to service condition.

19.4.2. Availability Target: the trains supplied shall achieve a minimum availability of 95%.

19.5. Availability Demonstration

19.5.1. The average availability of the trains shall be assessed after 18 months from the start of revenue operation with the first train supplied under the contract, in a specified train Maintenance Depot. The total maintenance down times for all trains shall be collected by the Engineer on monthly basis, and the average availability during the preceding six months, shall be worked out from the above formula.

19.5.2. In the event that the availability target is not achieved due to Rolling Stock, the determination of availability achievement in the preceding six month period shall be continued at monthly intervals until the target is achieved.

19.5.3. In the event that the availability target is not achieved, the Contractor shall, at his own expense, take whatever action is deemed necessary to meet the availability requirement.

19.6. Maintainability Requirements

19.6.1. Simplicity of maintenance, operation and emergency procedures, ease of repair of damaged cars and equipment, are most important. These together with ease of exterior and interior cleaning will be taken into account throughout the development of the design.

19.6.2. Particular attention shall be paid during the design of the cars to ensure that scheduled maintenance tasks are achieved in minimum time and using minimum manpower.

19.6.3. Those components, systems and assemblies which require routine maintenance, frequent attention or unit replacement, shall be easily accessible for in situ maintenance.

19.6.4. The Contractor shall develop a comprehensive maintenance programme for the trains.

19.6.5. The maintenance regime proposed for the train shall be developed during the design process. A Failure Mode Effect Analysis (FMEA) will be required, based on function and derived from the specification at conceptual design stage.

19.6.6. At pre-final design stage the Contractor will develop this FMEA to include required maintenance derived from each failure mode. Any other maintenance required for the train should be indicated at this stage.

19.6.7. The vehicle shall incorporate design, which reduces maintenance, substantially improving service intervals and component replacement. The design shall also minimize mean time to repair (MTTR) and costs throughout design life. MTTR is the ratio of cumulative time, including the access time expended during a time interval to the total number of relevant failures.

19.6.8. The objective of the maintainability program including corrective and preventive maintenance shall provide for:

(i) Enhancement of Vehicle availability.

(ii) Minimisation of maintenance cost.

(iii) Minimisation of vehicle down time.

19.6.9. During the design stage, the Contractor shall furnish a list of Least Replaceable Units (LRU's) for the equipments, Sub-system and Systems supplied, which should not take more than 30 minutes for replacement. Specific exceptions, if any, whose replacement is not achievable in 30 minutes, shall be indicated by the tenderers in their offer. In order to achieve this requirement, quick release connections such as plugs and adaptor shall be provided between LRU's and the equipment.

19.6.10. The Tenderer shall submit the expected MTTR of the identified key systems as listed in table 19.6, along with the bid.

Table 19.6: MTTR of major systems

S.N	System / Equipment	MTTR (hours) Target
(i)	Propulsion System	2.5
	a) Third rail current collector	1.5
	b) HSCB & Earthing switch	2.5
	c) Power Converter – Inverter	3.0
	d) Traction Motor	3.0
(ii)	Auxiliary Supply System	1.7
	a) Auxiliary Converter	2.0
	b) Battery Charger	2.0
	c) Back-up Batteries	1.0
(iii)	Air Supply and Friction Brake Equipment	2.0
(iv)	Door System and Controls	1.2
(v)	VAC System	1.5
(vi)	Communication System	1.2
(vii)	Couplers and Draft Gear	2.2
	a) Automatic couplers	2.2

S.N	System / Equipment	MTTR (hours) Target
	b) Semi permanent couplers	2.0
(viii)	Bogies	2.0
	a) Drive gear and coupling	2.0
	b) Primary suspension	2.0
	c) Mechanical Parts & Components	2.0
	d) Secondary suspension	2.0
(ix)	Lighting System	1.0
(x)	DMS	1.5

19.7. Maintainability Demonstration

19.7.1. The Contractor shall carry out tests to demonstrate that all maintainability predictions provided are met. All such tests shall be completed within six months after the delivery of first train. In the event that any maintainability target is not achieved, the Contractor shall at his own expense take whatever action is deemed necessary to meet the maintainability targets.

19.7.2. The Contractor shall ensure that all the required information including the related Maintenance Work Instructions (MWI) etc. are available before the commissioning of the first train into revenue service to enable him to demonstrate the maintainability targets.

19.8. Vehicle Maintenance Program

19.8.1. The Contractor shall submit to the Engineer a maintenance program, within the period specified in the delivery schedule, in order to reach the reliability and availability level defined in the performance requirements.

19.8.2. During the design stage, the Contractor shall submit downtime and manpower requirements for the maintenance inspections and service checks considered necessary for maintaining the trains under normal operational conditions as per table 19.7. The service check sessions shall include all routine maintenance activities including inspections, cleaning, washing, pest and rodent control etc.

Table 19.7: Maintenance Inspections and Service checks

Session	Interval (Minimum)	Manpower and downtime requirements (Maximum)	
		Downtime	Expected staff (man days)
Service Check 1			
Service Check 2, if any			
Service Check 3, if any			
……..			
Service Check n, if any			

19.8.3. The Contractor shall also submit periodicity, downtime and manpower requirements for the maintenance activities as listed in table 19.8, for maintaining the trains under normal operational conditions, during the design stage.

19.8.4. For achieving lowest operation and maintenance costs, the metro trains shall be designed according to the following criteria:

(i)	Design for a life cycle of	at least 35 years
(ii)	Main mid life overhaul after	15 years
(iii)	Interval for Revision / main inspection of traction and brake equipment and other safety relevant equipment	about 500 000 km

Table 19.8: Maintenance Activities

Type of activity	Frequency (days or km)	Expected staff (man days)	Downtime	Specific tools required
A				
B				
C				
…				
Half-life overhaul				

19.8.5. The Corrective Maintenance time as indicated above shall include defect identification, replacement of defective LRUs and restoration to service condition

19.8.6. For all the life of the metro train, this plan shall include also :

(i) maintenance concept (schedule, type of operation…),

(ii) maintenance documentation,

(iii) labour and infrastructure means,

(iv) tools and workshop equipment,

(v) spare parts (including wearing parts),

19.8.7. The trains shall operate with minimum attention between the specified inspection periods, and shall, under the operating conditions specified, operate between overhaul periods without requiring replacement of components other than those on the agreed list of consumable parts to be proposed by the Contractor and accepted by the Engineer.

19.8.8. Special tools shall be avoided wherever possible. If they are required, they shall be supplied by the Contractor in sufficient quantities to meet the maintenance requirements.

19.8.9. Equipment design shall be modular to minimise down time following failures of equipment and components. Provision for mechanical handling devices shall be provided for any single piece of equipment weighing more than 35kg. Equipment covers shall be provided with secure, visible, latching arrangements for easy inspection from the side of trains

19.8.10. All under frame equipment which cannot be handled manually shall be configured such that it can be removed and replaced from track level using fork lift trucks or lift tables, with recognition being given to the confined environment of the pit and the rail level and under frame dimensions. All under frame equipment shall be arranged such that it is capable of being removed and replaced without disturbing any other equipment.

19.8.11. If any equipment mounted above the ceiling requires the use of lifting equipment for its removal or refitting this shall be readily achievable without the risk of damage to the vehicle interior.

19.8.12. Removal and re-assembly of moving and wearing parts on bogies shall generally be carried out without the use of special tools.

19.8.13. Bogies shall be capable of being disconnected and reconnected to vehicle bodies with a minimum of operations. All connections must be easily and safely accessible to personnel located in pits or alongside the bogie at rail level. It shall be easy to inspect for correct reconnection, from alongside the bogie where possible. Preference will be given to a design which permits release of the bogie to permit the raising of the car body, without the need for a pit in the Lifting Berth.

19.8.14. Each vehicle shall be capable of being lifted complete with bogies without the need to attach extra restraints or supports for the bogies or wheels.

19.8.15. Lubrication points shall have button head type grease nipples, and shall be easily accessible from rail level and shall, where possible, be grouped together.

19.8.16. On-vehicle test equipment shall be used on a vehicle to discriminate between a fault on the main equipment and a fault on the control electronic equipment.

19.8.17. Should the electronic equipment be found to be faulty, the equipment shall enable fault finding to be carried out at module level.

19.8.18. Off vehicle test equipment shall be used in the depot repair centre. This equipment shall allow fault finding down to the smallest replaceable item of equipment.

19.8.19. The unit shall have equipment cases and modules that are connected to the main vehicle wiring via connectors which are proven in equivalent service duties to achieve high reliability and are easily removable in the event of equipment replacement.

19.8.20. Equipment to which access will be required for faultfinding shall be conveniently located. A list of such equipments and their location shall be supplied.

19.8.21. The unit shall have provision for the isolation and where applicable, earthing of all electrical sub-systems to facilitate safe and systematic maintenance and fault diagnosis.

19.8.22. It shall be physically impossible for plug and socket connections and connections on safety-critical circuits to be mismatched.

19.8.23. The unit shall have standard test points on pneumatic systems. There shall be unrestricted access to facilitate checks during routine maintenance and fault diagnosis.

19.8.24. The above mentioned features shall be suitable reflected in the respective design documents, as applicable, during the design stage.

19.9. Safety Assurance

19.9.1. Safety is defined as freedom from those conditions that can cause death, injury, occupational illness, or damage to or loss of equipment or property, or withdraw the train from service. Is considered as a risk all circumstance susceptible to cause injuries or person death (passengers, operation staff, maintenance staff), and by extension all event leading to a partial or total destruction of costly equipments. The objective of safety is expressed by"

the capacity of the rolling material to keep the physical integrity of the passengers and persons in general ". The safety of the metro train aims to reduce to an acceptable level the probability of occurrence of catastrophic and critical events (according to norm EN 50126-1).

19.9.2. The Contractor shall submit a 'System Safety Assurance Management Plan' for review and acceptance by the Engineer as specified in the Engineer's Requirements: General Specification chapter 2.

19.9.3. The System Safety Assurance Management Plan shall cover design, manufacture, testing, commissioning and integrated testing, and minimising the magnitude and seriousness of events or malfunctions, which could result in injury to patrons or staff and damage to equipment or property, but cannot be completely eliminated.

19.9.4. All equipment and systems, including software, affecting train safety and the safety of train crew and passengers, and/or identified as being "vital", shall be designed according to the following principles:

(i) Only components having a high reliability and predictable failure mode shall be used.

(ii) Components must be utilized in such a manner that ensures that a restrictive, rather than a permissive condition will result from a component failure. (For example: brakes will apply, rather than release; train will decelerate, rather than accelerate.).

(iii) Circuits shall be designed such that when a normally energized electric circuit is interrupted or de-energized, it will cause the controlled function to assume its most restrictive condition. (Broken wires, damaged or dirty contacts, a relay failing to respond when energized, etc., shall not result in an unsafe condition.)

(iv) System safety equipment design must be such that any single independent component or subsystem failure results in a restrictive condition. Failures that are not independent, those failures which, in turn, always cause others, must be considered in combination as a single failure and must not cause a permissive condition.

19.9.5. During the Design Review process, the Contractor shall submit analyses for Engineer review, which demonstrate compliance with these safety principles. These analyses shall address the following issues:

(i) Circuit design.

(ii) Hardware design (Failure Modes, Effect and Criticality Analysis).

(iii) Electrical interference.

(iv) Software errors.

(v) System failures.

19.10. Hazard Analysis report

19.10.1. Safety performance shall be assessed by the Risk levels of individual hazards identified during the System Assurance process.

19.10.2. The Contractor shall submit the Hazard Analysis report in accordance with the requirements of Chapter 2 of the Employer's Requirements – general Specification and shall evaluate and ensure that all the hazards are identified and satisfactorily resolved.

19.10.3. The Contractor shall take lead role in the interface Hazard Analysis for train borne equipment provided by other contractors.

19.10.4. The Contractor shall produce the Hazard Analysis Schedule for the complete train including all train borne systems and shall interface principally with the Signalling, Communication, Power Supply, Civil and Depot Contractor as well as any other Designated Contractors to obtain the information necessary, from their hazard analysis, to complete the analysis.

19.10.5. The Contractor shall, as part of the safety analysis, prepare analysis to identify Hazards and ensure their satisfactory resolution. The following analysis shall be prepared and submitted by the Contractor for the Engineer acceptance.

(i) Preliminary hazard analysis

(ii) Interface hazard analysis (excluding EMI)

(iii) Subsystem hazard analysis

(iv) Operating hazard analysis including maintenance

(v) Quantitative fault tree analysis

(vi) Failure modes effects and criticality analysis (FMECA)

19.10.6. The Hazard Analysis shall be carried out in accordance with EN50126-1 as the primary standard, or any other internationally accepted equivalent standard, in areas not adequately addressed by the former standard.

19.10.7. The Contractor shall compile a list of critical and catastrophic items identified as a result of hazard analysis, FMECA or by other means. The Contractor shall carryout the Hazard and FMECA for the following equipment / sub-systems / systems:

(i) Bogie and Suspension.

(ii) Vehicle Body.

(iii) Transmission Drive System.

(iv) Gangways.

(v) Coupler.

(vi) Brake System.

(vii) Door System.

(viii) VAC System.

(ix) Pneumatic System.

(x) Communication System.

(xi) HV and Propulsion System.

(xii) Auxiliary Power System.

(xiii) Control equipment.

(xiv) TMS.

19.10.8. All hazard resolution by procedural control shall be cross-referenced from the Critical and Catastrophic Items List to the appropriate manuals.

19.10.9. As example, and according to the probability of occurrence associated to the event, these objectives are The qualitative measures of hazard severity are defined as follows:

(i) Hazard Category I – Catastrophic: Operating conditions such that personnel errors, environment, design deficiencies, subsystem or component failure or procedural

deficiencies may cause death or system loss. The safety target shall be based on internationally accepted standards.

(ii) Hazard Category II – Critical: Operating conditions such that personnel errors, environment, design deficiencies, subsystem or component failure or procedural deficiencies may cause severe injury to personnel, severe occupational illness or major system damage. The safety target for the occurrence of all Category II hazards summed together shall again be based on internationally accepted standards.

(iii) Hazard Category III – Marginal: Operating conditions such that personnel errors, environment, design deficiencies, subsystem or component failure or procedural deficiencies, may cause minor injury to personnel, minor occupational illness or minor system damage.

(iv) Hazard Category IV – Negligible: Operating conditions such that personnel errors, environment, design deficiencies, subsystem or component failure or procedural deficiencies will not result in injury to personnel occupational illness or damage to the system.

(v) The Contractor shall submit a Schedule for Hazard Analysis Submissions within 30 days of Notice to Proceed(NTP) and the Preliminary Hazard Analysis shall be submitted within 6 months of Notice to Procedd(NTP). This draft shall include a comprehensive assessment of potential equipment failure modes during normal operating and overload conditions and assess the performance of the equipment for a range of hazard conditions. The final draft shall be submitted by the completion date of final design.

19.10.10. The procedures for Operation, Maintenance, Training and the Contractor's Quality Assurance manuals shall incorporate resolution of hazards so identified from this hazard analysis. Proper cross-referencing to the hazards and resolution measures shall be provided in all these aforementioned documents.

19.10.11. The following targets norms shall be employed for the Fault Tree Analysis These norms are subject to review by the Engineer during the detailed design stage, and mutually agreed upon.

(i) No single point failure shall lead to fatality.

(ii) No combination of undetected failure and double point failures shall result in fatality.

(iii) No combination of undetected failure and single point failure shall result in major injury.

19.10.12. Source of all failure rates employed to be indicated in the Hazard Analysis.

19.10.13. All hazard analyses submitted to the Engineer are to be standardized by the Contractor such that format and forms employed by all sub-contractors are the same.

19.11. Fire Safety Analysis Report

19.11.1. The Contractor shall prepare a Fire Safety analysis report for review and acceptance by the Engineer as required in chapter 3 of the Employer's Requirements - General Specification. Materials used in the cars shall conform to fire safety requirements of BS 6853: 1999 and NF F 16 101, NF F 16 102 or the latest edition of other equivalent international standards, subject to the acceptance of the Engineer.

19.11.2. Whichever Standard is selected for meeting the Fire Safety Criteria, then that standard shall be declared, and once accepted by the Engineer, its requirements shall be met consistently throughout.